CSDS 611 – Advanced topics and methods of Data Science I

Assignment 3

Due: Saturday, October 31, by 8:00PM

Problem: We consider a Gaussian process

$$f \sim \mathsf{GP}_{\mathbb{R}}(\mu, C)$$

with continuously differentiable mean μ and the squared exponential covariance kernel C.

Show that the derivative of f is also a Gaussian process and find its mean and covariance kernel.